Day 2 — Task 1: Git Basics

Create, commit and push hello.py to GitHub

The task

Create a small Python file **hello.py** in your repository with simple code (for example, printing your name and city). Then practise these commands in order:

```
git status
git add .
git status
git commit -m "Add hello program"
git push
```

Then modify hello.py by adding one more print statement and run the same commands again.

Key terms

Term	Meaning
Repository (repo)	A project folder tracked by Git; the history lives in the hidden .git folder
Untracked	Git sees the file but is not tracking it yet
Staging area	The list of changes ready to go into the next commit (filled by git add)
Commit	A saved snapshot of the project in your local repo, with a message
Branch	A line of work; the default is main (older Git uses master)
Remote / origin	The online copy of the repo (GitHub); origin is its usual name
Push	Upload local commits to the remote

The flow

```
Working folder --add--> Staging area --commit--> Local repo --push--> GitHub
(untracked/modified)      (ready)                  (saved)             (online)
```

What I did — step by step

1	`git init` Turned the GIT_exercise folder into a Git repository. Git created the hidden .git folder and started on a branch called **master**.
2	`Created hello.py → git status` Git listed hello.py under **Untracked files** and said “No commits yet” because the repository was brand new.
3	`git add .` Moved hello.py into the **staging area**. The dot means “everything in this folder”.
4	`git status` The file now showed under **Changes to be committed**, confirming the add worked.
5	`git commit -m "add hello program"` Saved a snapshot in the local repository. Output: *1 file changed, 1 insertion(+)* — one file with one line of code.
6	`git push → failed` *fatal: No configured push destination.* The local repo did not know which GitHub repository to send to. A repo made with git init has no remote until you add one.
7	`git branch -M main` Renamed the branch from master to **main**, the name GitHub uses by default.
8	`git remote add origin … → errors` The URL ended up on a separate line, so PowerShell tried to run it as a command (“term is not recognized”). The saved URL was also mistyped as **day2.g** instead of **day2.git**, so the push said *Repository not found*. Running git remote add origin again gave *remote origin already exists* because the wrong URL was already saved under that name.
9	`git remote set-url origin https://github.com/Deepikakanchi/day2.git` The correct fix: **set-url** changes the URL of an existing remote instead of adding a new one.
10	`git remote -v` Showed the fetch and push URLs, confirming both were now correct.
11	`git push -u origin main → success` Uploaded the commit to GitHub and created the main branch there (*[new branch] main -> main*). The **-u** links local main to origin/main, so plain **git push** works from now on.

Final working command sequence

```
git init
git status
git add .
git status
git commit -m "add hello program"
git branch -M main
git remote add origin https://github.com/Deepikakanchi/day2.git
git remote -v
git push -u origin main
```

Next part — modify and push again

Add one more line to hello.py, for example:

```
print("My name is <your name>")
print("I am learning Git")      # new line
```

Then run:

```
git status                                # hello.py shows as "modified"
git add .
git status                                # now under "Changes to be committed"
git commit -m "Add another print statement"
git push                                  # works without -u origin main now
```

Refresh the GitHub repo: the new line appears in hello.py and the history shows two commits.

Errors and fixes

Error	Cause	Fix
No configured push destination	No remote added	git remote add origin <url>
Repository not found	Wrong URL (typo) or repo not created on GitHub yet	Check the URL, then git remote set-url origin <url>
remote origin already exists	Tried to add origin twice	Use git remote set-url instead of add
“term is not recognized” (PowerShell)	URL typed as its own command	Keep the whole command on one line

Command cheat sheet

Command	What it does
git init	Start a new repository in the current folder
git status	Show untracked, modified and staged files
git add . / git add hello.py	Stage all changes / one file
git commit -m "msg"	Save staged changes as a snapshot
git log --oneline	Show commit history briefly
git branch -M main	Rename the current branch to main
git remote add origin <url>	Connect the local repo to GitHub
git remote set-url origin <url>	Change the remote URL
git remote -v	Show remote URLs
git push -u origin main	First push; sets upstream tracking
git push	Push later commits

⚠ **Remember / common mistakes**

- Create the repository on GitHub **before** pushing, and copy its URL exactly (it ends in .git).
- -u is only needed on the first push of a branch.
- Run git status often — it tells you what to do next.